Driessen, Lin, and Phalippou (2012) estimate betas of 1.3 for buyout funds and approximately 3 for VC funds, consistent with Korteweg and Sorensen (2010). Driessen, Lin, and Phalippou find that between 1980 and 2003—the period where the nonreturn IRR, multiple, and PME measures are highest (see Figures 18.3–18.5)—buyout funds underperform the market by 0.4% per month. The underperformance of VC funds after adjusting for market risk is 1.1% per month. This is large underperformance. After incorporating additional size and value-growth factors, the alphas for buyout and VC funds are –0.7% and –1.0% per month, respectively. These are very large underperformance numbers.

The size and value-growth exposures are important sources of risk premiums for buyout funds. Phalippou (2013) shows that any performance advantage for buyout funds, if it exists, is attributable to buying small or even tiny companies with a value bias. Both these tilts give buyout fund portfolios higher returns than the market, but you could do the same with cheap index funds. Oh, and buyout funds use lots of leverage.

Another perspective on the value added by PE is provided by Jegadeesh, Kräussl, and Pollet (2009). They examine *listed* PE funds and funds-of-funds. They report excess returns of –7% to –5% per year after controlling for risk factors in public equity markets. They also find significant exposures of listed PE to size and value factors.

Other Risk Factors

Illiquidity is an important risk factor in PE returns. Franzoni, Nowak, and Phalippou (2012) show that PE investments are exposed to the *same* illiquidity risk factor as public equity. That is, if an asset owner wants to collect an illiquidity risk premium, she might be better off collecting it in public markets rather than private markets (see also chapter 13). Franzoni, Nowak, and Phalippou find that the illiquidity risk premium accounts for approximately 3% of PE returns, which in their sample are 18%. That is, the remaining 15 percentage points of PE returns are due to other risk factors. Alphas are already modest (or even negative) before controlling for illiquidity risk and become even smaller after illiquidity risk is taken into account. Franzoni, Nowak, and Phalippou's estimate of PE alpha is zero to three decimal places after capturing illiquidity, size, and value effects.

Robinson and Sensoy (2011b) document that VC funds become "liquidity sinks" when valuations are low. Both LP capital contributions and LP distributions are pro-cyclical when liquidity is high. PE funds provide liquidity during booms, when market valuations are high, but there is no liquidity when valuations are low. This was made clear in 2007–2008 during the financial crisis as PE commitments became an albatross for many investors forced to meet capital calls at a time they most needed to conserve cash.[26]

[26] Harvard University and CalPERS were two such investors. See "Liquidating Harvard," Columbia CaseWorks ID #100312 and "California Dreamin': The Mess at CalPERS," Columbia CaseWorks, #120306.